Franck-Hertz Experiment

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1 Introduction

The Franck-Hertz experiment provides the first direct experimental evidence for the quantization of atomic energy levels, confirming theoretical predictions initially made by the Bohr model. By determining the exact energy required to excite a specific atomic transition in mercury, the experimental can be used to calculate an experimental value for Planck's constant, h.

When free electrons are accelerated through mercury vapour, they undergo collisions with the mercury atoms. If the kinetic energy of an incident electron is less than the energy difference between the atom's ground state and its lowest accessible excited state (ΔE), the collisions are perfectly elastic. In this regime, there is no reduction in the electron's kinetic energy. However, if the electron's kinetic energy is greater than or equal to ΔE, inelastic collisions can occur. During an inelastic collision, the electron transfers a discrete quantum of kinetic energy to the mercury atom, exciting it to a higher energy state, and continues its path with the surplus energy. By observing the energy lost by electrons undergoing these inelastic collisions, the excitation energy ΔE of the atom can be accurately measured. This energy difference relates to the accelerating voltage (V) by

$$\Delta E = eV \tag{1}$$

where e is the elementary charge.

The specific atomic transition measured in this setup corresponds to the excitation of mercury's $6s6p^3P_1$ state. Unlike lower energy transitions, this specific state decays rapidly back to the ground state by emitting an ultraviolet photon with a wavelength of $\lambda = 253.7$ nm. Using the measured excitation energy and the known wavelength of the emitted photon, Planck's

constant can be derived using the de Broglie relation for photon energy:

$$\Delta E = hf = \frac{hc}{\lambda} \tag{2}$$

We can then substitute ΔE from Equation 1 into Equation 2 and rearrange to solve for h:

$$h = \frac{\Delta E \lambda}{c} = \frac{eV\lambda}{c} \tag{3}$$

2 Methods

Experimental Setup

The experimental setup centers around a completely evacuated glass tube containing a drop of mercury, which is enclosed within a thermostatically controlled oven to precisely regulate the mercury vapour pressure. The internal components of the tube operate similarly to a conventional vacuum tube, with a filament, a cathode, a grid-like anode, and a collector. The filament is powered by a 6.3 VAC power supply, which heats the cathode to emit electrons via thermionic emission. These electrons are then accelerated toward the grounded anode by a variable 0–50 V DC power supply. Because the anode is grounded, the filament and cathode are configured to float with respect to ground, which is necessary for the current measurement. A picoammeter, capable of reading currents between 10×10^{-9} A and 10×10^{-7} A, is connected to measure the fraction of electrons that reach the collector. A 1.5 V battery is placed between the anode and the collector to provide a fixed negative retarding potential; this ensures that electrons arriving at the anode with less than 1.5 eV of kinetic energy are turned back. The collector connections are completely shielded to prevent externally induced currents, and the battery is carefully insulated from ground to stop the 1.5 V potential from driving an unwanted leakage current through the sensitive picoammeter. The ambient temperature of the mercury vapour is monitored and adjusted using an OMEGA CN370 temperature controller, which reads data from a thermocouple attached to a metal ring around the glass tube.

Experimental Procedure

To begin the experiment, the initial temperature on the OMEGA CN370 controller is set approximately 5 °C below the actual target temperature before turning on the power switch. The optimal operating temperature

to observe the maximum number of current peaks is typically established between 150 °C and 180 °C, and care is taken to ensure the system never exceeds the safety limit of 250 °C. Once the temperature stabilizes, the anode voltage is gradually increased from zero while observing the picoammeter, which must display a current in the negative direction. Initially, the accelerating voltage is kept below 20 V to safely locate the current peaks without overloading the tube. Pushing the voltage too high risks forming a plasma discharge inside the tube. If a plasma discharge occurs, it acts as a low resistance pathway, causing the 1.5 V battery to drive a large positive current through the picoammeter; should this happen, the procedure requires the anode voltage to be immediately turned down to zero before resuming. By systematically varying the accelerating voltage and recording the corresponding collector current, at least four distinct peaks are identified. This process yields the necessary data to plot the collector current against the anode voltage, allowing for the precise measurement of the voltage separation between successive peaks.

Data Analysis

The data is collected using an Agilent DSO1012A oscilloscope, which captures the collector current signal as a function of time during a linear voltage sweep of the anode from 0 to 39 V. The oscilloscope data is then exported to a CSV file, which is processed using Python. Because the data is recorded over a timescale rather than a voltage scale, the Python script, presented in Listing 1 first converts the time axis to voltage using the known sweep rate of the anode voltage. To find the excitation states, the script scans the signal to identify the precise anode voltage separation (δV) between these adjacent peaks and determines the associated standard error. Finally, it uses this average separation to calculate an experimental value and propagated error for Planck's constant using Equation 3, and generates an annotated plot of the collector current versus anode voltage to visually verify the results.

Listing 1: Python script for analyzing time-voltage oscilloscope data to determine Planck's constant.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path
from scipy.signal import find_peaks
import scipy.constants as const
from uncertainties import ufloat
from typing import Union
```

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# Constants
LAMBDA_M = 253.7e-9
V_PP = 39.0
V_F = 6.6

def load_and_preprocess_data(data_path: Path) -> pd.DataFrame:
    """
    Load the Franck-Hertz experiment data and isolate a single sweep.

    :param data_path: Path to the CSV file containing the experimental data.
    :return: A pandas DataFrame containing the isolated sweep.
    """
    df = pd.read_csv(data_path, header=0, usecols=[0, 1], names=['Time', 'Collector_Signal'])
    df['Signal_Diff'] = df['Collector_Signal'].diff()
    reset_indices = df.index[df['Signal_Diff'] < -30].tolist()

    if len(reset_indices) >= 2:
        return df.iloc[reset_indices[0] + 1:reset_indices[1]].copy()
    return df.copy()

def map_time_to_voltage(sweep: pd.DataFrame, v_pp: float) -> pd.DataFrame:
    """
    Map the time axis to an anode voltage axis using a linear assumption.

    :param sweep: DataFrame containing the sweep data.
    :param v_pp: Peak-to-peak voltage.
    :return: DataFrame with the mapped anode voltage.
    """
    t_min = sweep['Time'].min()
    t_max = sweep['Time'].max()
    sweep['Anode_Voltage'] = ((sweep['Time'] - t_min) / (t_max - t_min)) * v_pp
    return sweep

def detect_peaks(sweep: pd.DataFrame) -> tuple[np.ndarray, np.ndarray]:
    """
    Detect prominent peaks in the collector signal.

    :param sweep: DataFrame containing the sweep data with mapped anode voltage.
    :return: A tuple containing an array of peak voltages and
```

```python
        an array of peak signals.
    """
    peaks, _ = find_peaks(sweep['Collector_Signal'], prominence=5, distance=3)
    peak_voltages = sweep['Anode_Voltage'].iloc[peaks].values
    peak_signals = sweep['Collector_Signal'].iloc[peaks].values
    return peak_voltages, peak_signals

def calculate_results(peak_voltages: np.ndarray, lambda_m: float) -> None:
    """
    Calculate and print the physical results including Planck's constant using scipy.constants.

    :param peak_voltages: Array of detected peak voltages.
    :param lambda_m: Wavelength of the mercury resonance line.
    """
    delta_vs = np.diff(peak_voltages)
    avg_delta_v = np.mean(delta_vs)
    std_err_delta_v = np.std(delta_vs, ddof=1) / np.sqrt(len(delta_vs))

    delta_v_u = ufloat(avg_delta_v, std_err_delta_v)

    h_exp_u = (const.e * delta_v_u * lambda_m) / const.c
    h_exp_val = h_exp_u.n
    h_exp_err = h_exp_u.s

    percent_error = (abs(h_exp_val - const.h) / const.h) * 100

    print(f"Average Voltage Separation (dV) = {avg_delta_v:.3f} V +/ {std_err_delta_v:.3f} V")
    print(f"Energy of the excited state ~= {avg_delta_v:.3f} eV")
    print(f"\nCalculated Planck's Constant (h) = {h_exp_val:.4e} +/- {h_exp_err:.4e} J * s")
    print(f"Accepted Planck's Constant (h)   = {const.h:.4e} J * s")
    print(f"Percent Error                    = {percent_error:.2f}%")

def plot_results(sweep: pd.DataFrame, peak_voltages: np.ndarray, peak_signals: np.ndarray, savefig: Union[str, Path]) -> None:
    """
    Plot the collector signal versus anode voltage with detected peaks highlighted.

    :param sweep: DataFrame containing the sweep data.
```

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    :param peak_voltages: Array of detected peak voltages.
    :param peak_signals: Array of detected peak signals.
    :param savefig: Path to save the generated plot.
    """
    plt.figure(figsize=(10, 6))
    plt.plot(sweep['Anode_Voltage'], sweep['Collector_Signal'], label='Collector Current', color='blue')
    plt.plot(peak_voltages, peak_signals, "rx", markersize=10, label='Identified Peaks')

    plt.title('Franck-Hertz Experiment: Collector Current vs. Anode Voltage', fontsize=14)
    plt.xlabel('Anode Voltage (V)', fontsize=12)
    plt.ylabel('Collector Signal (Relative)', fontsize=12)
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.legend()
    plt.tight_layout()
    plt.savefig(savefig, bbox_inches='tight', dpi=300)
    plt.show()

def plot_raw_data(df: pd.DataFrame, savefig: Union[str, Path]) -> None:
    """
    Plot the raw collector signal versus time.

    :param df: DataFrame containing the raw experimental data.
    :param savefig: Path to save the generated plot.
    """
    plt.figure(figsize=(10, 6))
    plt.plot(df['Time'], df['Collector_Signal'], label='Raw Collector Signal', color='green')
    plt.title('Franck-Hertz Experiment: Raw Collector Signal vs. Time', fontsize=14)
    plt.xlabel('Time (s)', fontsize=12)
    plt.ylabel('Collector Signal (Relative)', fontsize=12)
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.legend()
    plt.tight_layout()
    plt.savefig(savefig, bbox_inches='tight', dpi=300)
    plt.show()

def main() -> None:
    """
    Main execution function for Franck-Hertz data analysis.
    """
    data_path = Path(__file__).parent.parent / "data" / "franck_hertz_data.csv"
    out_path = Path(__file__).parent.parent / "out"
    out_path.mkdir(exist_ok=True, parents=True)
```

```python

    raw_data = pd.read_csv(data_path, header=0, usecols=[0, 1], names=['Time', 'Collector_Signal'])
    plot_raw_data(raw_data, savefig=out_path / "raw_data_plot.png")

    sweep = load_and_preprocess_data(data_path)
    sweep = map_time_to_voltage(sweep, V_PP)

    peak_voltages, peak_signals = detect_peaks(sweep)
    print(f"Identified {len(peak_voltages)} peaks at Anode Voltages: {np.round(peak_voltages, 2)} V\n")

    calculate_results(peak_voltages, LAMBDA_M)
    plot_results(sweep, peak_voltages, peak_signals, savefig=out_path / "franck_hertz_plot.png")

if __name__ == "__main__":
    main()
```

3 Raw Data

The raw data collected from the oscilloscope is presented in Figure 1. The plot shows the collector current as a function of the anode voltage, with distinct peaks corresponding to the inelastic collisions of electrons with mercury atoms. The voltage separation between these peaks is crucial for determining the excitation energy of the mercury atoms and, subsequently, Planck's constant.

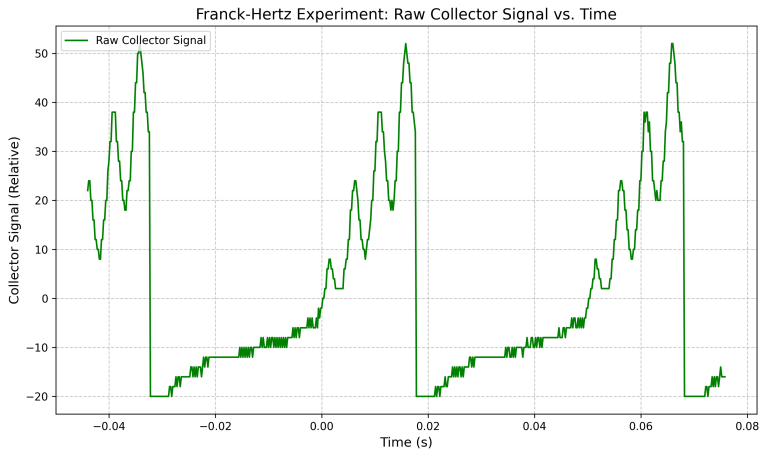

Figure 1: Collector current versus anode voltage from the oscilloscope data. The peaks indicate the inelastic collisions of electrons with mercury atoms.

4 Analysis and Results

Figure 2 shows the processed data, where the collector current is plotted against the anode voltage.

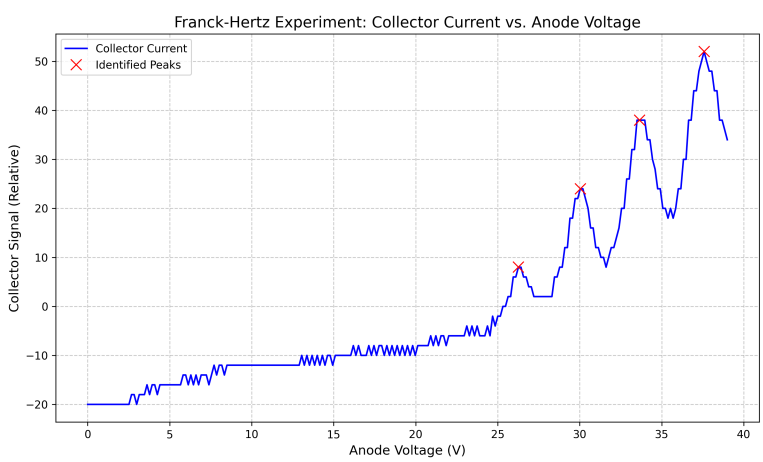

Figure 2: Plot of collector current versus anode voltage, showing the distinct peaks corresponding to the excitation of mercury atoms.

The experimental data yielded four distinct local maxima in the collector

current as the anode voltage was swept. These peaks occurred at accelerating voltages of approximately 26.26 V, 30.04 V, 33.65 V, and 37.58 V.

By taking the differences between adjacent peaks, the average voltage separation was calculated to be $\Delta V = 3.77 \pm 0.09$ V. This separation directly corresponds to the discrete kinetic energy lost by an electron during a single inelastic collision with a mercury atom, providing the measured excitation energy $\Delta E \approx 3.77$ eV of the atom.

Using this excitation energy and the known emission wavelength of the unexcited state, we can substitute into Equation 3 to arrive at an experimental value for Planck's constant, $h = (5.12 \pm 0.12) \times 10^{-35}$ J s.

5 Discussion

The periodic local maxima observed in the collector current successfully demonstrate the quantization of atomic energy levels. The average voltage separation of (3.77 ± 0.09) V indicates that electrons lose a discrete amount of kinetic energy during inelastic collisions with mercury atoms, which corresponds to the excitation energy of the atom.

However, it is physically significant that the level detected in this apparatus is actually mercury's second exited state, characterized by the atomic structure $6s6p^3P_1$. There exists a lower energy state ($6s6p^3P_0$), but it is not observed in this experiment because it decays via a forbidden transition. Because it cannot rapidly decay via photon emission, this lower state is metastable and possesses a long lifetime within the gas, primarily decaying only through physical collisions with the walls of the tube. Under continuous electron bombardment, the population of this metastable state builds up until the rate of electron collisions that excite it perfectly balances the rate of collisions that de-excite it. Consequently, this lower state plays no role in the average energy balance of the electrons and remains completely undetected by this experimental setup. Instead, the measured 3.77 eV transition corresponds entirely to the $6s6p^3P_1$ state, which decays rapidly back to the ground state by emitting the 253.7 nm ultraviolet photon, making the atom immediately available to undergo another excitation cycle.

When compared to the accepted value of Planck's constant, $h = 6.626 \times 10^{-34}$ J s, our experimental value of $h = (5.12 \pm 0.12) \times 10^{-35}$ J s is significantly lower, with a percent error of approximately 22.7%. This discrepancy can be attributed to several sources of error in the experiment.

First, the derivation assumes that the kinetic energy of the electrons is entirely determined by the accelerating voltage. In reality, the electrons

are emitted via thermionic emission and possess a small, non-zero initial thermal energy from the heated cathode. This intrinsic thermal spread causes a broadening of the electron energy distribution, making the current drops less sharply defined and shifting the apparent position of the peaks.

Secondly, the experimental value of the anode voltage at the peaks is influenced by the work functions of the distinct metals used for the cathode and anode. While taking the voltage difference between successive peaks – rather than absolute values – largely mitigates this contact potential difference, any localized variations in the grid's geometry or surface potentials can still introduce systematic shifts in the peak-to-peak distances.

Finally, the inherent probability cross-sections of the collisions themselves contribute to the error. An electron does not necessarily undergo an inelastic collision the exact moment its kinetic energy reaches 3.77 eV; it must physically encounter an available mercury atom. This means electrons often travel slightly further and gain kinetic energy slightly above the exact threshold before a collision actually occurs, smearing the distances between the theoretical and observed peaks.

6 Conclusion

The Franck-Hertz experiment successfully demonstrated the quantization of atomic energy levels by measuring the discrete energy lost by electrons through mercury vapour. The average voltage separation between collector current peaks was measured at $\delta V = (3.77 \pm 0.09)\,\mathrm{V}$, corresponding to an excitation energy of roughly 3.77 eV for mercury's $6s6p^3P_1$ state.

Applying the de Broglie relation to this measured excitation energy yielded an experimental value for Planck's constant of $h = (5.12 \pm 0.12) \times 10^{-35}\,\mathrm{J\,s}$. While this calculation carries a 22.7% error compared to the accepted value — primarily due to the thermal energy spread of the emitted electrons and contact potential differences between the electrodes — the observation of uniform, discrete energy transfers rigorously confirms the theoretical prediction of quantized atomic energy states.